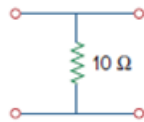


Problem

For the single-element two-port network in Fig. 19.64(a), **B** is:

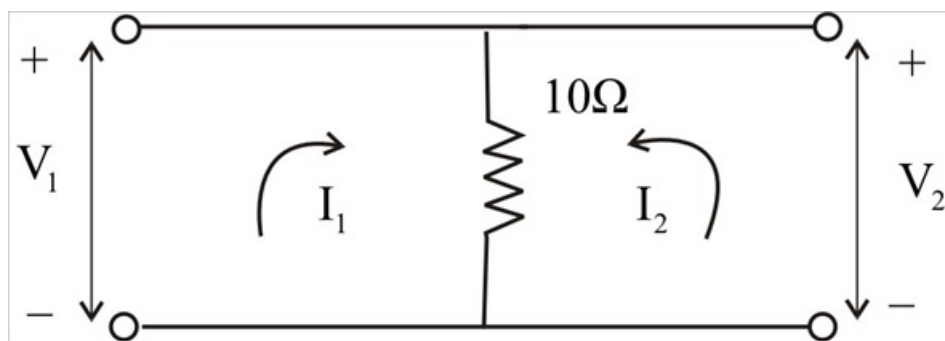
- (a) 0
- (b) 5
- (c) 10
- (d) 20
- (e) undefined

Figure 19.64(a)



Step-by-step solution

Step 1 of 2



Step 2 of 2

We have

$$10I_1 + 10I_2 = V_1$$

$$10I_1 + 10I_2 = V_2$$

$$B = -\left. \frac{V_1}{I_2} \right|_{V_2=0} \quad [V_2 = 0 \Rightarrow I_1 = -I_2]$$

$$= -\frac{(10I_1 + 10I_2)}{I_2}$$

$$= -\frac{(10I_1 - 10I_1)}{I_2}$$

$$= 0$$